

Understanding QKD performance under classical-signal coexistence

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M5 Assignment 5.1 | September 2026

2110

Experimental observations

98.8%

Variance retained by PC1–PC2

39.2%

Held-out LDA accuracy

Research question and main contribution

How do attenuation, quantum bit error rate (QBER), and secure key rate (SKR) vary together, and how well do these measurements distinguish five classical-signal conditions? This analysis combines unsupervised structure discovery, supervised classification, and multivariate regression. These methods answer different questions: a compact performance pattern need not accurately identify the condition that produced it.

PCA summarizes joint variation; K-means and hierarchical clustering describe operating regimes; LDA evaluates recovery of known labels; and a two-response regression estimates attenuation and condition associations jointly. All reported numbers, tables, and figures are generated by the accompanying R script.

Dataset and scope

The source is Horgan et al.'s version-2 Zenodo dataset, DOI 10.5281/zenodo.21792844 [1]. Five CSV files contain baseline measurements and classical-signal injections of 3, 7, 9, and 12 dBm. This subset concerns one QKD system; it is not an analysis of every device in the broader record. Restricting added fibre to 0 km holds that recorded factor fixed. It does not balance attenuation across conditions or establish a causal power effect.

Interdisciplinary connection

The interpretation connects quantum communications with reliability engineering and statistical decision-making. PCA and clustering summarize multivariate operating states, while the confusion matrix exposes the limits of diagnosing an experimental condition from observed performance. In a monitoring application, confusing conditions and overlooking low-throughput states have different consequences. A defensible decision process examines class-specific errors and usable throughput together rather than relying on one accuracy number.

1 Data preprocessing and exploratory analysis

1.1 Variables, validation, and transformations

Table 1: Analysis variables and their roles.

Variable	Type / unit	Role and treatment
NoiseCondition	Five-level factor	LDA classes; factor predictor in regression.
Sweep_km	Numeric / km	Fixed at zero; excluded as a constant predictor.
VOA_dB	Numeric / dB	Attenuation predictor.
QBER	Numeric / fraction	Error rate; validated in $[0, 1]$.
SecureKeyRate_bps	Numeric / bits s ⁻¹	Nonnegative throughput; zero is valid.
LogSKR	Numeric / log scale	$\log_{10}(\text{SKR} + 1)$, replacing raw SKR in multivariate methods.

The script checks required columns, numeric types, finite values, permitted labels, sample sizes, and nonzero predictor variability before analysis. There are 0 missing values in the supplied dataset, so no imputation or case deletion is needed. Missing or invalid required values cause an explicit error rather than silently changing the analysis sample. Repeated equal readings are retained because identical values do not establish duplicate records; the selected files do not provide unique run identifiers.

The log transform reduces SKR's scale disparity while retaining zero-throughput observations. Raw SKR and its transform are not both included as PCA predictors, which would count the same quantity twice. PCA and clustering use standardized variables. LDA scaling is estimated on training data only. Regression preserves response units and centers attenuation to make its intercept interpretable.

Table 2: Condition-level descriptive statistics. Pooled summaries reflect each condition's observed attenuation distribution.

Condition	n	Mean QBER	Median SKR (bps)	Zero-SKR (%)
Baseline	471	0.0484	3,541	8.7
3 dBm	400	0.0493	5,673	13.0
7 dBm	354	0.0467	8,400	8.2
9 dBm	319	0.0491	3,319	9.4
12 dBm	566	0.0457	91,610	22.1

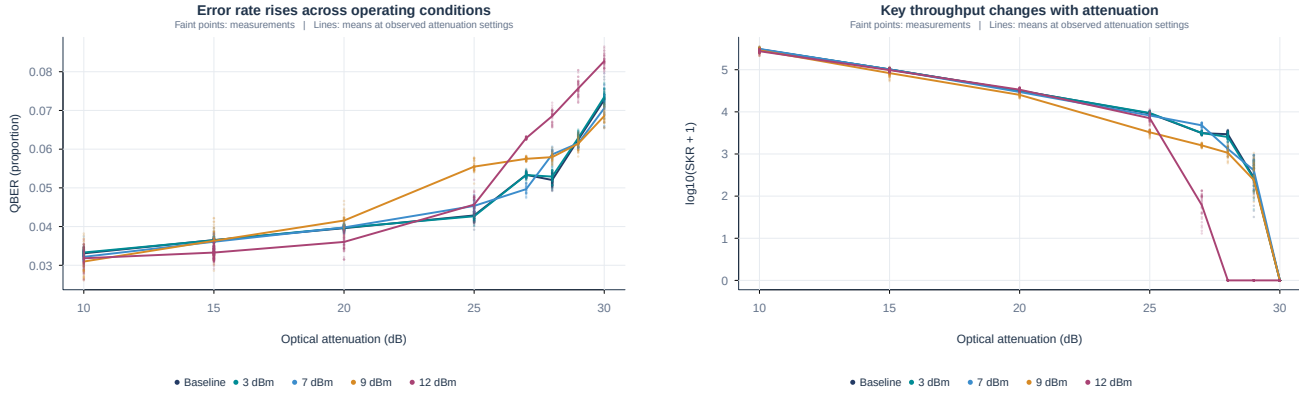
The analysis contains 2110 measurements across five conditions, with group sizes ranging from 319 to 566. Attenuation spans 10.0–30.0 dB, QBER spans 0.0261–0.0865, and raw SKR spans 0–354598 bits/s. There are 277 zero-SKR measurements (13.1%); these are retained, and the $\log_{10}(\text{SKR} + 1)$ transformation preserves their value at zero.

The strongest absolute pairwise correlation is between LogSKR and QBER ($r = -0.957$). In particular, attenuation correlates 0.878 with QBER and -0.824 with transformed SKR. These pooled associations combine within-condition trends and differences in the sampled condition/attenuation mix; they do not by themselves isolate a causal effect of injected power.

For example, mean sampled attenuation is 23.13 dB for baseline and 19.61 dB for 12 dBm, while their median raw SKR values are 3541 and 91610 bits/s, respectively. Those medians summarize different attenuation distributions. A raw median comparison must therefore not be interpreted as a controlled injected-power effect.

At the shared attenuation of 28 dB, mean SKR is 2969 bits/s for baseline and 0 bits/s for 12 dBm. The 12 dBm condition also has 22.1% zero-throughput readings overall. This matched-setting comparison explains why its high pooled median is not evidence of better performance under stronger injection.

1.2 Attenuation-dependent patterns



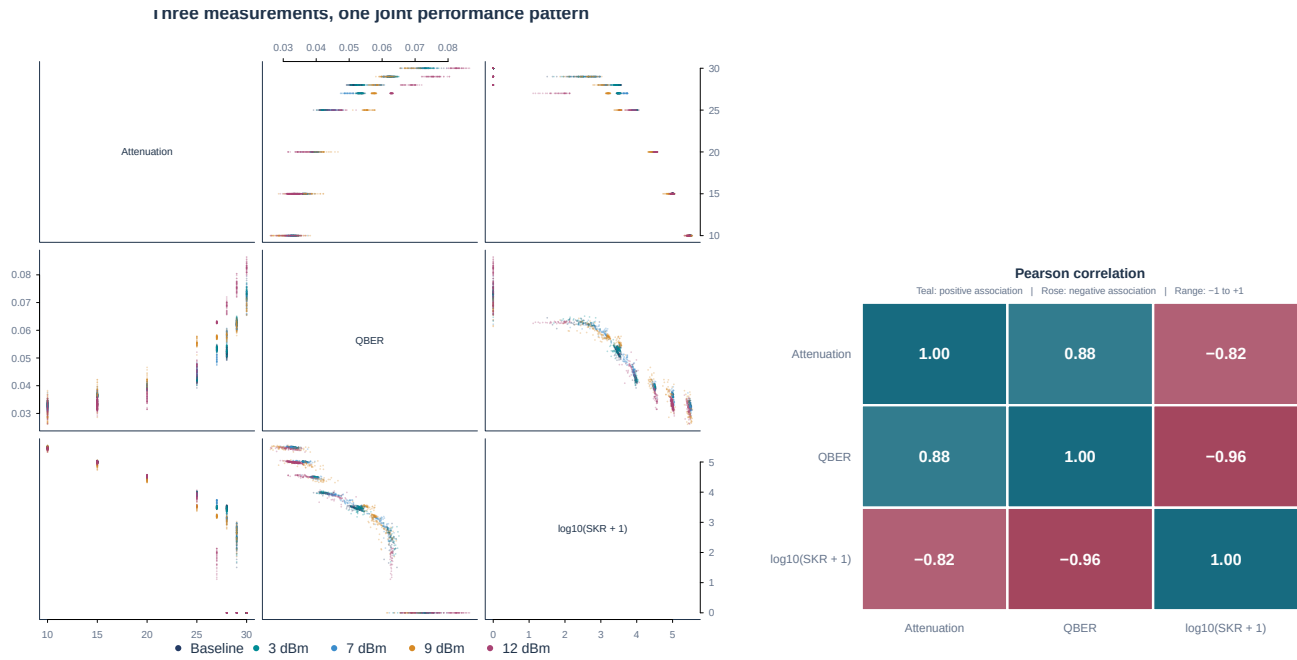
(a) Error-rate response.

(b) Key-throughput response.

Figure 1: Measurements and condition-specific mean trends. Compare conditions at shared attenuation settings; pooled medians alone can reverse the apparent condition ordering.

Figure 1 distinguishes within-condition attenuation trends from between-condition differences. Near key-generation collapse, a moderate change in QBER can accompany a large throughput loss. Lines connecting measured setting means are descriptive summaries, not fitted physical laws or uncertainty intervals.

Strong dependence motivates dimension reduction, but correlation does not establish causation. Attenuation is a design variable as well as a predictor, so the multivariate pattern partly reflects how the experiment was sampled.



(a) Scatterplot matrix.

(b) Pearson correlations.

Figure 2: Pooled pairwise dependence. Discrete measurement settings and nonlinearity remain visible in the scatterplots.

2 Principal component analysis

2.1 Method and component selection

The R function `prcomp(..., center=TRUE, scale.=TRUE)` applies PCA to attenuation, QBER, and LogSKR [2]. If $\mathbf{Z}$ is the standardized matrix, scores are $\mathbf{t}_j = \mathbf{Z}\mathbf{w}_j$, with orthogonal loading vectors ordered by explained variance. Scaling prevents the predictor with the largest numerical units from dominating.

The retention rule is the smallest number of components reaching at least 80% cumulative explained variance, supported by the scree plot. It selects 1 component(s). A two-dimensional biplot remains useful for inspecting the smaller second direction and condition overlap; plotting two axes does not mean that two are required by the retention rule. PC1 retains 92.5% of standardized variance, PC2 retains 6.3%,

Table 3: Eigenvalues and explained variance of all principal components.

Component	Eigenvalue	Variance (%)	Cumulative (%)
PC1	2.774	92.5	92.5
PC2	0.190	6.3	98.8
PC3	0.036	1.2	100.0

and their cumulative share is 98.8%. The 80% retention rule selects 1 component(s); the two-dimensional display is still useful for inspecting the smaller second-direction contrast.

The displayed PC1 loadings are VOA_dB +0.561, QBER +0.591, LogSKR -0.580; PC2 loadings are VOA_dB +0.808, QBER -0.242, LogSKR +0.537. Variables with the same loading sign move together along a component, whereas opposite signs define a contrast. The largest absolute contributions are

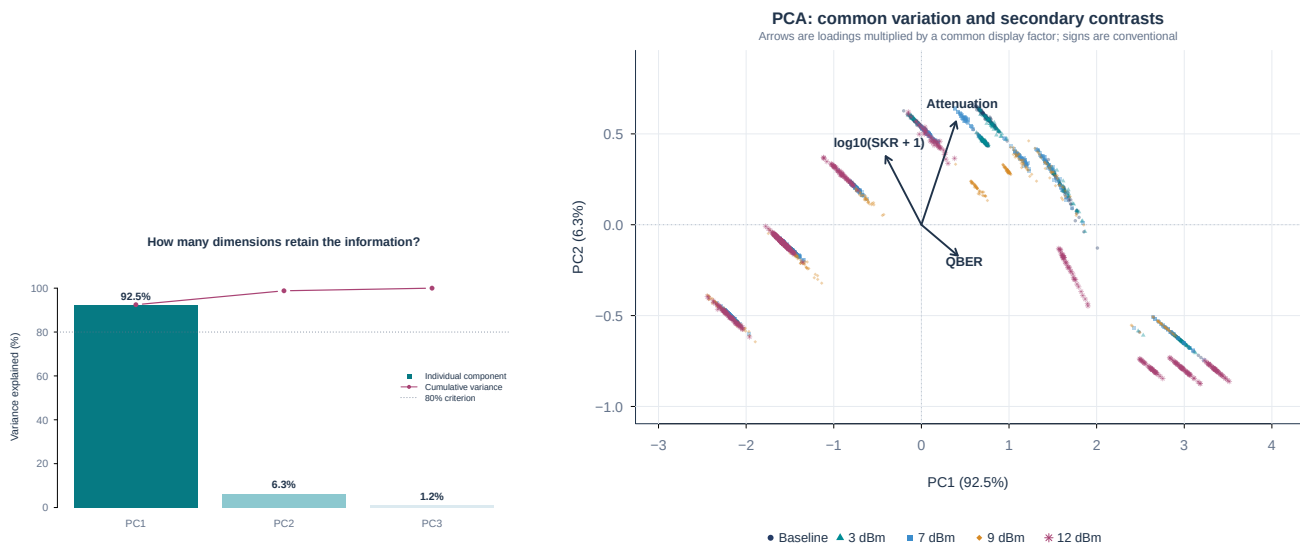
Table 4: PCA loading vectors. A simultaneous sign reversal of a component and its scores leaves the solution unchanged.

Variable	PC1	PC2	PC3
VOA_dB	0.561	0.808	-0.177
QBER	0.591	-0.242	0.770
LogSKR	-0.580	0.537	0.613

QBER on PC1 and VOA_dB on PC2. Component signs are arbitrary and have been oriented so each largest loading is positive; reversing an entire component changes no statistical conclusion.

Higher PC1 scores combine higher Attenuation, QBER with lower $\log_{10}(\text{SKR} + 1)$. Higher PC2 scores combine higher Attenuation, $\log_{10}(\text{SKR} + 1)$ with lower QBER.

PC1 can therefore be interpreted as an attenuation/error-throughput deterioration axis: it contrasts greater attenuation and error with lower key throughput. PC2 mainly captures the attenuation contrast left after that shared direction is removed. The near-one-dimensional pattern explains why the variables can be compressed, while the smaller contrasts remain relevant for distinguishing conditions.



(a) Scree and retention evidence.

(b) Scores and loading directions.

Figure 3: PCA describes covariance without using class labels. Loading arrows are rescaled for visibility; their displayed lengths cannot be compared directly with distances between observations.

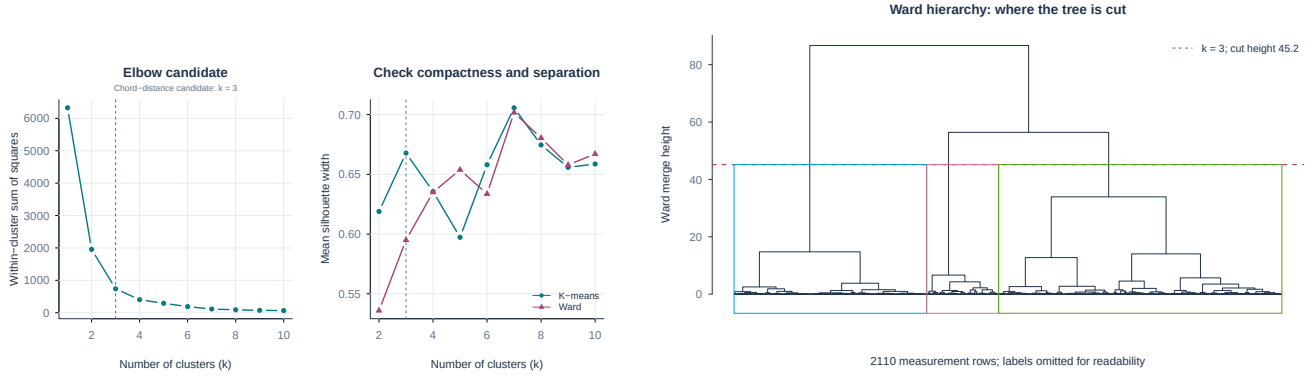
Explained variance measures reconstruction, not classification accuracy. A low-variance direction may still matter for distinguishing groups or rare operating states. LDA therefore retains all three predictors rather than automatically discarding the smaller PCs.

3 Cluster analysis

3.1 K-means and Ward hierarchical clustering

Both methods use the three standardized variables with Euclidean geometry. K-means minimizes squared distance to centroids and uses multiple random starts to reduce initialization sensitivity. Ward's `ward.D2` method joins groups to limit the increase in within-cluster variation. Neither fit uses condition labels.

K-means candidates span $k = 1, \dots, 10$. The elbow estimate uses the largest deviation from the chord joining the first and last within-cluster sum-of-squares values. The dendrogram and separation diagnostics assess whether the selected cut describes useful structure. A selected k is a modeling choice, not proof of a unique natural number of groups.



(a) K-means selection evidence.

(b) Ward hierarchy and selected cut.

Figure 4: Cluster-number assessment: diminishing returns in the elbow plot and larger merges in the hierarchy provide complementary evidence.

The geometric elbow selects the candidate $k = 3$. At this cut, mean silhouette width is 0.668 for the final K-means solution and 0.595 for Ward clustering. Across the tested cuts, the largest mean silhouettes occur at $k = 7$ (K-means) and $k = 7$ (Ward). The elbow is a compact-summary heuristic, not proof of a uniquely optimal number of physical regimes.

The Ward dendrogram cut lies between merge heights 33.92 and 56.42. The adjusted Rand index comparing K-means and Ward memberships is 0.650 (1 denotes identical partitions; approximately 0 denotes chance-level agreement). Both methods use the same standardized variables, so agreement is descriptive and does not establish independent validation.

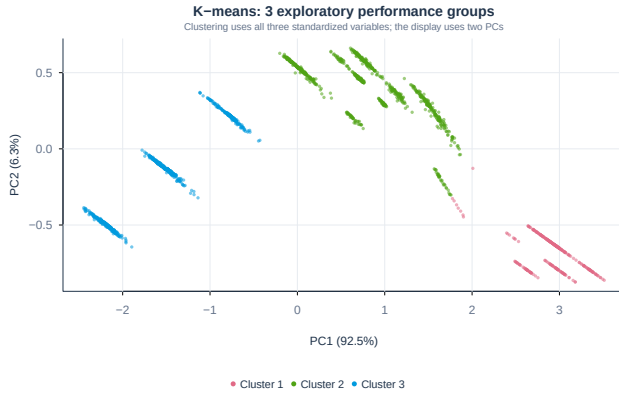
Cluster 1 ($n = 285$) has mean attenuation 29.48 dB, mean QBER 0.0731, mean log-SKR 0.04, and 97.2% zero SKR; its largest condition share is 12 dBm (46.3%). Cluster 2 ($n = 856$) has mean attenuation 27.08 dB, mean QBER 0.0536, mean log-SKR 3.30, and 0.0% zero SKR; its largest condition share is Baseline (31.0%). Cluster 3 ($n = 969$) has mean attenuation 14.64 dB, mean QBER 0.0350, mean log-SKR 5.02, and 0.0% zero SKR; its largest condition share is 12 dBm (36.9%).

Every fitted cluster contains observations from all five experimental conditions. The clusters therefore describe near-collapse, reduced-throughput, and higher-throughput performance regimes rather than recovering the injection labels. Labels are ordered by increasing mean log-SKR, so their numeric ordering follows throughput in this solution.

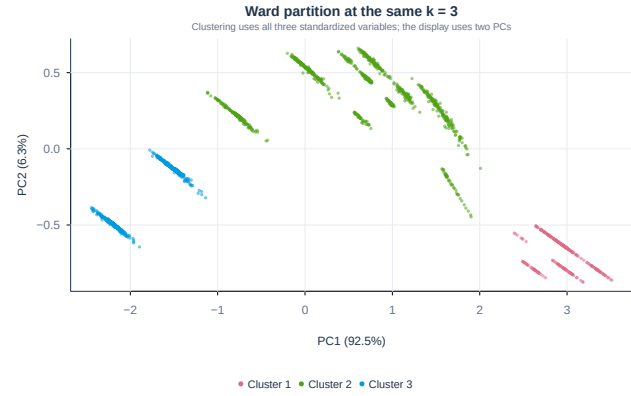
Clustering describes regions of performance space, not causal mechanisms or validated alarm states. Attenuation produces visible bands, so some separation reflects experimental design. Reading profiles and condition cross-tabulations together avoids equating an unsupervised partition with the five injection labels.

Table 5: K-means profiles in interpretable units. Cluster labels identify a fitted partition, not known experimental conditions.

Cluster	n	Mean VOA	Mean QBER	Mean log-SKR	Zero SKR (%)
1	285	29.48	0.0731	0.04	97.2
2	856	27.08	0.0536	3.30	0.0
3	969	14.64	0.0350	5.02	0.0



(a) K-means partition.



(b) Ward partition at the same k .

Figure 5: Partitions projected onto the same PCA axes. Fits use all three standardized variables; label numbers are arbitrary within each method.

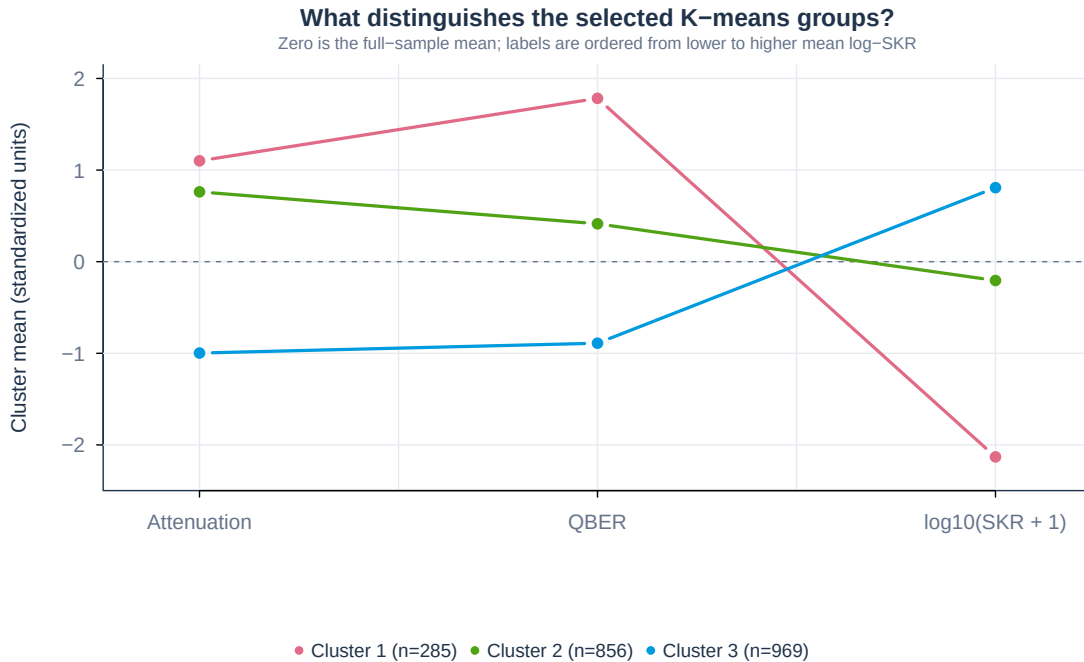


Figure 6: Standardized profiles show which features characterize each operating regime. Zero is the full-sample mean, not a physical safety threshold.

4 Linear discriminant analysis

4.1 Groups, discriminant functions, and scores

The target is the five-level injection condition; predictors are attenuation, QBER, and LogSKR. A seeded, stratified 70/30 split yields 1475 training observations and 635 test observations. Training means and standard deviations scale both partitions. `MASS::lda()` uses training class proportions as priors; the untouched test partition supplies the classification evaluation [3].

LDA finds directions with large between-group separation relative to pooled within-group variation. Three predictors and five classes permit at most three discriminant functions:

$$LD_j = a_{j1}z_{VOA} + a_{j2}z_{QBER} + a_{j3}z_{LogSKR}.$$

Coefficient magnitudes indicate conditional contributions within each standardized direction. With correlated predictors, they are not independent or causal importance measures. Reversing an entire function's sign does not change its meaning.

Table 6: Discriminant coefficients for the standardized predictors.

Predictor	LD1	LD2	LD3
VOA_dB	0.907	1.929	0.005
QBER	2.456	-3.406	-0.488
LogSKR	3.236	-1.730	0.528

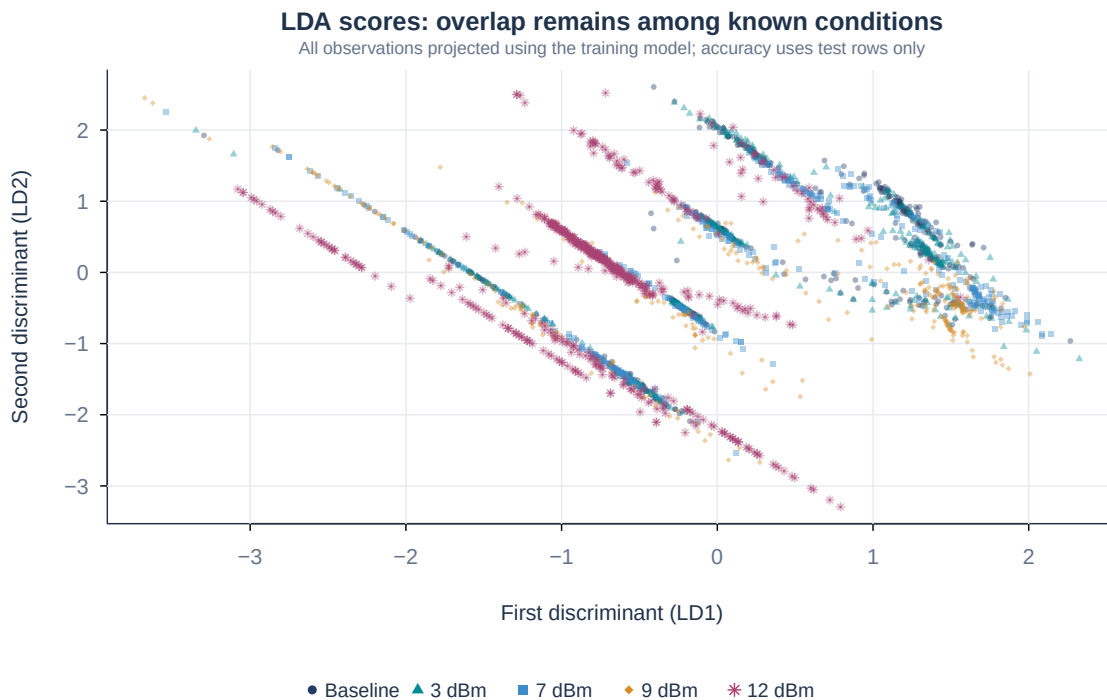


Figure 7: LDA score visualization. Group overlap is interpreted alongside held-out classification results rather than treated as an accuracy estimate.

4.2 Performance and difficult groups

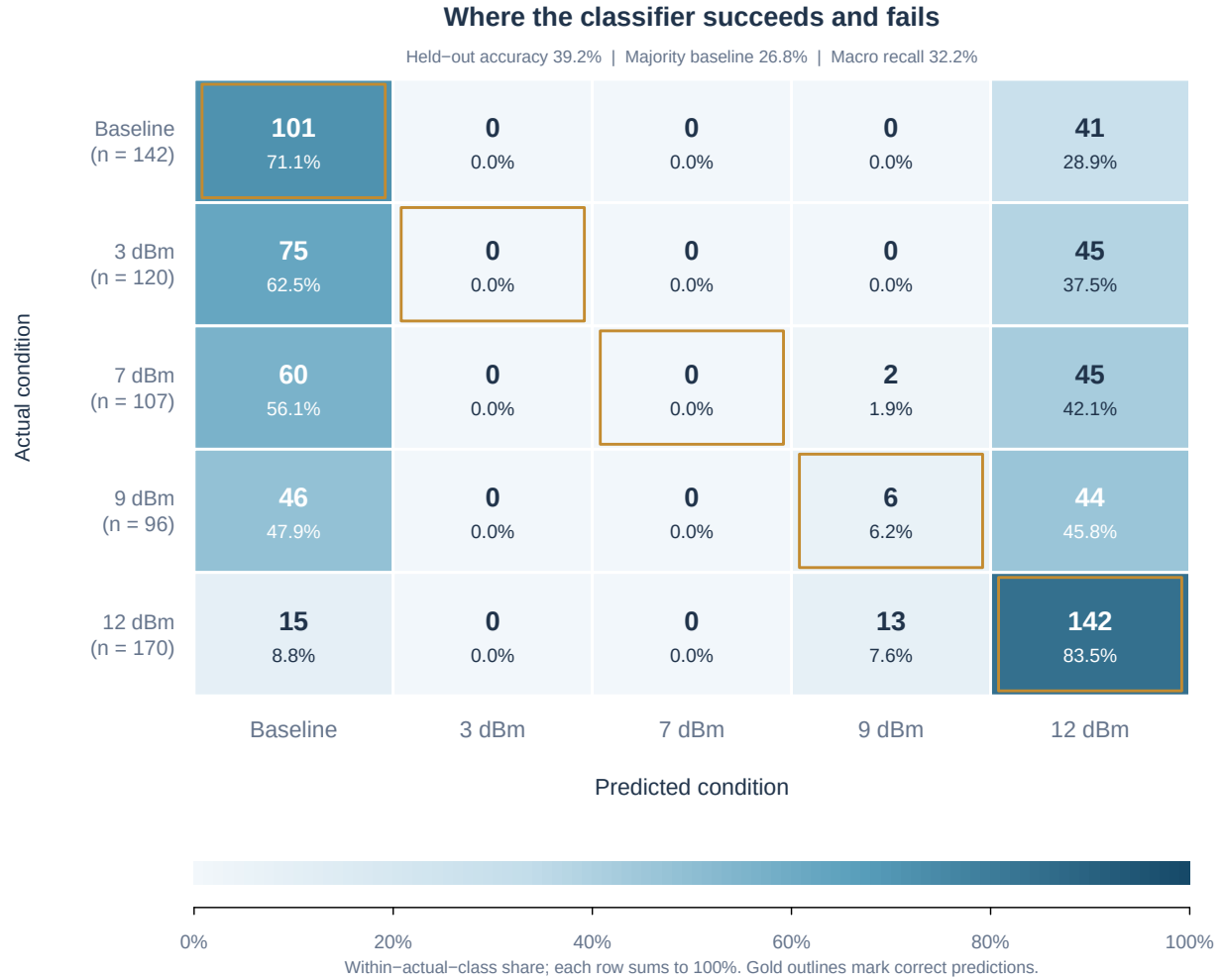


Figure 8: Held-out confusion matrix. Cells contain counts and percentages of their actual-class row. The diagonal shows correct predictions; off-diagonal cells identify specific errors.

Table 7: Class-specific evaluation. Recall is the fraction of an actual group recovered; precision is the fraction of a predicted group that is correct. Unavailable precision is shown explicitly for an unpredicted class.

Condition	Test n	Predicted n	Recall (%)	Precision (%)
Baseline	142	297	71.1	34.0
3 dBm	120	0	0.0	–
7 dBm	107	0	0.0	–
9 dBm	96	21	6.2	28.6
12 dBm	170	317	83.5	44.8

The fixed stratified split uses 1475 training and 635 test rows. LDA correctly classifies 249 of 635 test observations, giving accuracy 39.2%. Always predicting the training-majority class (12 dBm) gives 26.8% on the same test rows, so the accuracy difference is 12.4 percentage points. Macro recall is 32.2%, which weights the five conditions equally.

Class recall ranges from 0.0% to 83.5%; the lowest recall includes 3 dBm (0 correctly classified out of 120). Each heatmap cell shows a count and its percentage of the actual class, and the gold diagonal marks correct predictions. Off-diagonal mass therefore makes the model's errors visible even when class sizes differ. No test observations were predicted as 3 dBm or 7 dBm; precision is therefore undefined for those classes.

These estimates concern held-out rows from the same campaign. Repeated measurements at shared settings can occur in both partitions; this is not validation on independent runs or unseen operating conditions. LDA's common within-class covariance and approximately Gaussian predictor assumptions are also approximations for bounded QBER and zero-heavy SKR.

The largest absolute standardized coefficient in LD1 is LogSKR (+3.236), and in LD2 it is QBER (-3.406). These identify the strongest conditional contributions to the two plotted directions, not independent causal effects. Because attenuation, error, and throughput are strongly correlated, substantial coefficients can coexist with poor class separation.

Baseline has recall 71.1%; its most frequent error is prediction as 12 dBm (41 of 142 actual observations, 28.9%). 3 dBm has recall 0.0%; its most frequent error is prediction as Baseline (75 of 120 actual observations, 62.5%). 7 dBm has recall 0.0%; its most frequent error is prediction as Baseline (60 of 107 actual observations, 56.1%). 9 dBm has recall 6.2%; its most frequent error is prediction as Baseline (46 of 96 actual observations, 47.9%). 12 dBm has recall 83.5%; its most frequent error is prediction as Baseline (15 of 170 actual observations, 8.8%).

Row normalization makes errors comparable despite unequal class sizes. Accuracy is interpreted against the majority-class baseline and macro-average recall, which gives each actual class equal weight. Exceeding the baseline can coexist with failure to recover multiple groups. Those failures are substantive evidence about overlap and model suitability, not a reason to suppress results.

4.3 Validation limits

The split samples rows from the same campaign. Similar measurements at shared settings can occur in both partitions, and independent run identifiers are absent. Performance therefore describes within-campaign held-out observations, not new devices, runs, or fibre lengths. LDA's common-covariance and approximately Gaussian class assumptions are imperfect for curved trajectories, zero-SKR pileups, and unequal spreads. Future work should hold out complete runs or operating settings and compare more flexible classifiers before operational use.

5 Multivariate regression analysis

5.1 A model with two quantitative responses

The rubric includes multivariate regression in addition to the task list's LDA. To address both, this section fits a genuine multivariate multiple linear regression:

$$\begin{bmatrix} \text{QBER}_i & \text{LogSKR}_i \end{bmatrix} = \mathbf{x}_i^T \mathbf{B} + \varepsilon_i.$$

The predictor vector includes an intercept, centered attenuation, four condition indicators with baseline as reference, and their attenuation interactions. The R model is

```
lm(cbind(QBER, LogSKR) ~ VOA_c * NoiseCondition).
```

Condition coefficients compare groups at the common centering value. Interactions describe slope

differences from baseline. Responses keep their own units, so a QBER coefficient is not directly comparable in magnitude with a LogSKR coefficient.

Outcome-specific coefficients and fit summaries are complemented by joint tests using Pillai's trace [4]. These tests concern the response vector rather than a single outcome. Their interpretation remains conditional on the linear model and error assumptions. Multivariate multiple regression jointly models

Table 8: Outcome-specific regression fit. In-sample R^2 is not held-out prediction accuracy.

Outcome	R^2	Adjusted R^2	RMSE	Residual df
QBER	0.816	0.815	0.0061	2100
Log-SKR	0.748	0.747	0.8500	2100

Table 9: Joint multivariate tests with Pillai's trace. The generated interpretation identifies the tested model terms or comparisons.

Sequential term	Pillai	Approx. F	df	Reference p
Attenuation (centered)	0.808	4410.95	2, 2099	< 0.001
Noise condition	0.161	46.00	8, 4200	< 0.001
Attenuation $\times$ condition	0.168	48.26	8, 4200	< 0.001

Table 10: Coefficient estimates and standard errors for both responses. Baseline is the reference; attenuation is centered.

Term	QBER est.	QBER SE	Log-SKR est.	Log-SKR SE
Baseline at mean attenuation	0.0463	0.0003	3.905	0.040
Baseline slope per dB	0.0015	0.0000	-0.170	0.006
At center: 3 dBm	0.0009	0.0004	-0.120	0.059
At center: 7 dBm	0.0005	0.0004	-0.060	0.061
At center: 9 dBm	0.0027	0.0004	-0.254	0.062
At center: 12 dBm	0.0046	0.0004	-0.909	0.055
Extra slope: 3 dBm	0.0002	0.0001	-0.022	0.009
Extra slope: 7 dBm	0.0000	0.0001	-0.000	0.009
Extra slope: 9 dBm	0.0002	0.0001	-0.012	0.009
Extra slope: 12 dBm	0.0010	0.0001	-0.116	0.008

QBER and log-SKR using attenuation centered at 21.69 dB, five noise-condition levels, and their interaction. Baseline is the reference condition. At the center, its fitted QBER is 0.0463 and fitted log-SKR is 3.905; its slopes per additional dB are +0.0015 and -0.170, respectively. Condition coefficients describe differences at the center, and interaction coefficients describe differences from the baseline slopes.

The within-sample R^2 values are 0.816 for QBER and 0.748 for log-SKR. For the attenuation-by-condition interaction, the sequential Pillai statistic is 0.168, with approximate $F = 48.26$ and reference $p < 0.001$. The displayed MANOVA tests are sequential in formula order; with an interaction present, main-effect coefficients must be interpreted at the reference condition or centered attenuation.

The residual-versus-fitted and normal Q-Q plots assess curvature, unequal spread, and non-normal tails. Among observed rows, 0 fitted QBER values fall outside $[0, 1]$ and 0 fitted log-SKR values fall below

zero, indicating whether the unconstrained linear model crosses physical boundaries. Standard errors and reference tests assume an error structure that repeated within-setting measurements may violate. The regression is therefore a descriptive association model for this campaign, not evidence of causation or confirmed population-level significance.

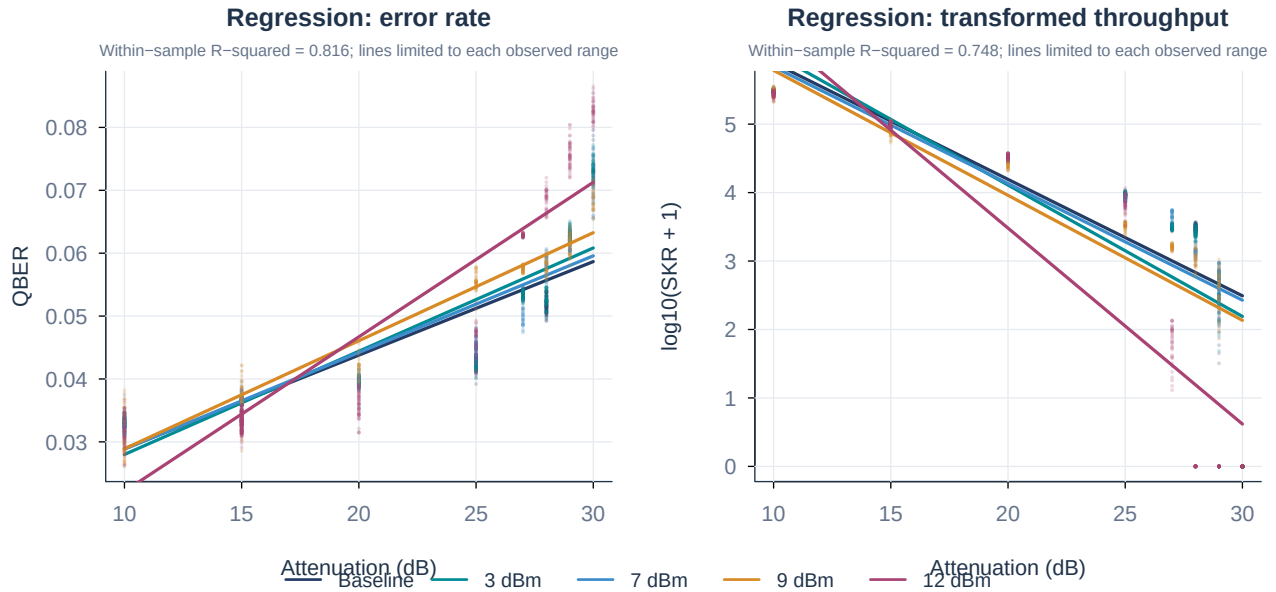


Figure 9: Observed performance and regression fits over attenuation. Lines describe modeled associations within the observed domain, not physical response laws or extrapolation guarantees.

The linear form is an interpretable starting point, not a physical model of key-generation collapse. Residual curvature signals missing structure, changing spread challenges constant variance, and quantile-plot departures challenge normal-error inference. QBER is bounded and LogSKR has a mass at zero. Repeated within-setting measurements also weaken independent-error assumptions. Nominal standard errors and p -values are therefore exploratory. Run-level replication, flexible attenuation effects, and explicit modeling of zero throughput would strengthen follow-up analysis.

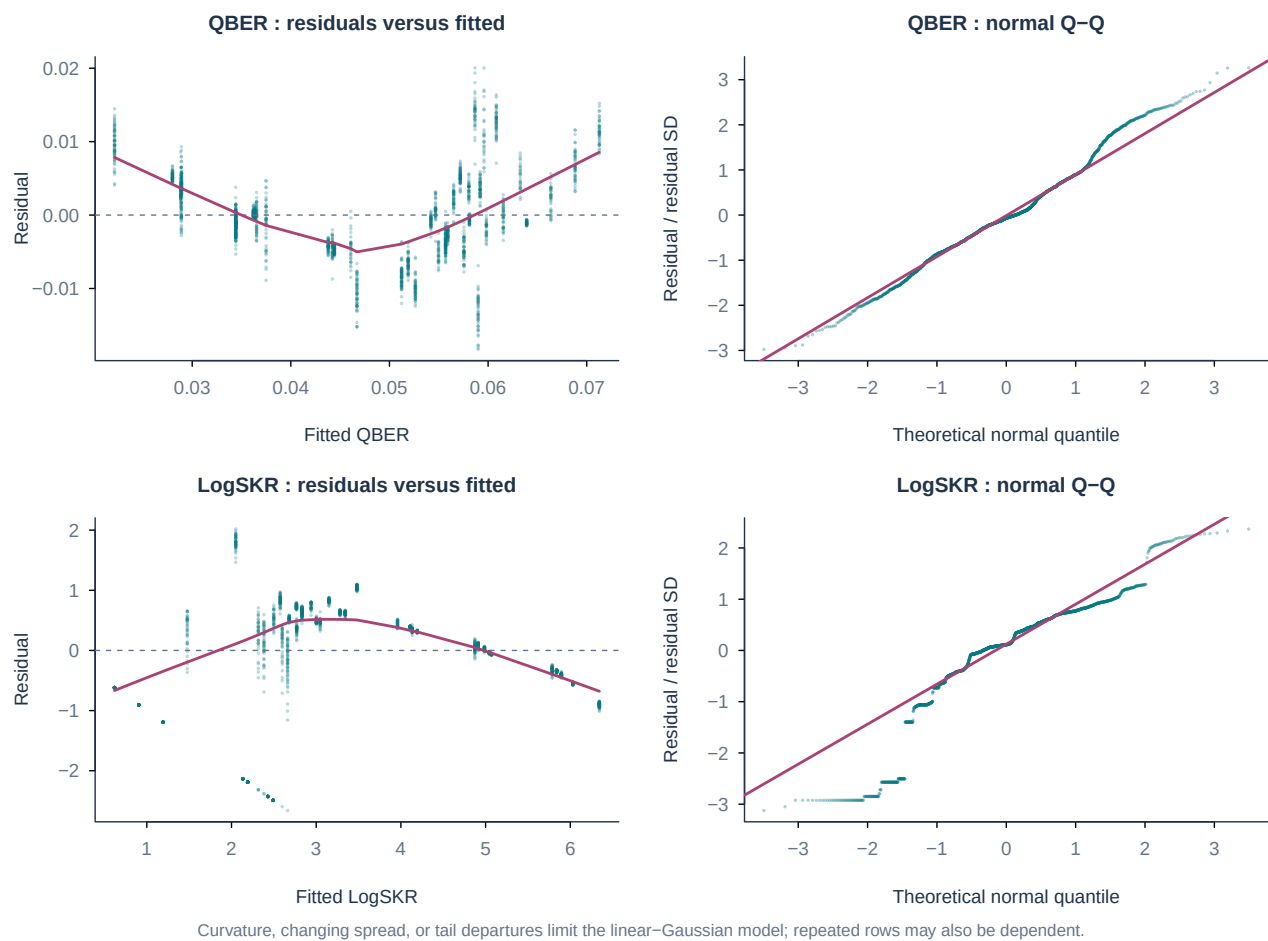


Figure 10: Regression diagnostics assess the linear, constant-variance, Gaussian-error approximation.

6 Synthesis, ethics, and conclusion

6.1 What the methods jointly establish

PCA finds a compact representation: 1 component(s) meet the 80% rule, while two explain 98.8% of standardized variance. K-means selects 3 clusters at the chosen level of detail, with profiles describing operating regimes. These findings concern performance geometry; they do not imply that five injection labels are separable.

LDA achieves 39.2% held-out accuracy, with its error pattern providing the more informative diagnostic. Regression instead estimates joint response associations with attenuation and condition, including slope differences. Together, these methods distinguish compression, grouping, label prediction, and response modeling rather than treating them as equivalent measures of success.

6.2 Reliability and responsible interpretation

Joint changes in error rate and key throughput can guide candidate operating-state descriptions in reliability engineering. Such descriptions require physical validation and explicit decision thresholds before becoming alarms. Low recall for a class cautions against inferring an injection label confidently from these measurements alone.

Ethical practice includes preserving CC BY 4.0 attribution, retaining legitimate zero-throughput readings, documenting transformations, and reporting weak group-level performance. Neither a low QBER nor a fitted cluster certifies cryptographic security. The conclusions apply to the selected system and 0-km campaign; other devices, fibre lengths, and operational scenarios require additional evidence.

6.3 Conclusion

The data support substantial dimension reduction and interpretable performance regimes, but recovering every condition label is harder. A small set of correlated measurements can summarize performance well while providing limited diagnostic specificity. Multivariate regression adds a joint account of error and throughput, with nonlinearity and inference limits made explicit. The analysis offers a reproducible statistical foundation for better experimental design and validation.

Reproducibility and submission

The R script reads the submitted combined CSV when available, or uses the five original CSVs. It validates input, creates every result and figure, records seeds and split membership, and saves `sessionInfo()`. GitHub Actions reruns the analysis and compiles the report. The submission set contains:

- M5Assign 5.1 Data__Ronggon_Asif.R
- M5Assign 5.1 Data__Ronggon_Asif.csv
- M5Assign 5.1 Report__Ronggon_Asif.pdf

References

- [1] J. Horgan, N. Quigley, B. Hajar, D. Briantcev, A. Kaszubowska-Anandarajah, and D. Kilper. *QKD Network Characterisation and Coexistence Datasets with Standardised Performance Metrics and Experimental Methods*, v2. Zenodo, 2026. <https://doi.org/10.5281/zenodo.21792844>.
- [2] R Core Team. *Principal Components Analysis: prcomp*. R documentation. <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/prcomp.html>.
- [3] W. N. Venables and B. D. Ripley. *Modern Applied Statistics with S*, 4th ed. Springer, 2002. MASS documentation: <https://stat.ethz.ch/R-manual/R-devel/library/MASS/html/lda.html>.
- [4] R Core Team. *Summary Method for Multivariate Analysis of Variance*. R documentation. <https://stat.ethz.ch/R-manual/R-devel/library/stats/html/summary.manova.html>.

A Assignment coverage

Table 11: Coverage of the rubric and explicit LDA tasks. This crosswalk is not a predicted grade.

Criterion	Evidence
Preprocessing and visualization (20)	Section 1: source, variable roles, descriptive statistics, missing-value policy, transformations, scaling, scatterplots, correlations, and attenuation trends.
PCA and interpretation (20)	Section 2: standardized PCA, every variance proportion, retention rule, scree plot, biplot, loading signs, and reduction implications.
Clustering and interpretation (20)	Section 3: K-means and Ward methods, elbow and dendrogram assessment, projected clusters, profiles, and partition comparison.
Multivariate regression (20)	Section 5: two responses, coefficients and interactions, joint tests, fit, visualizations, diagnostics, and limitations.
Clarity and insights (20)	Structured research question, cited source, numerical interpretation, method synthesis, reliability connection, ethics, and conclusion.
LDA (task list)	Section 4: predefined classes and predictors, MASS fit, coefficients, scores and group plot, held-out confusion matrix and accuracy, and difficult-class interpretation.
Submission components	.R script, external dataset .csv, and PDF with the required naming pattern.

The task list explicitly requests LDA while the rubric names multivariate regression. Both are included so neither written requirement is omitted. The instructor determines how that discrepancy is graded.